Structural Damage Recognition Using Transfer Learning

Course: UE24CS352A — Machine Learning (Sec-C, Team C-22, Project 31)
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Abstract— Rapid post-disaster structural integrity assessment is crucial for life safety and emergency resource allocation. Manual inspection is hazardous, subjective, and slow. We develop an automated computer vision pipeline for binary structural damage recognition (Damaged vs. Undamaged) on the benchmark PEER Hub ImageNet (Φ-Net) Task 2 dataset. Tailored for consumer hardware constraints, our approach employs an ImageNet-pretrained MobileNetV2 backbone (include_top=False) with disk-backed bottleneck feature caching. Evaluated on the official held-out test set of 1,460 images, the model achieves **84.25% test accuracy**, an **ROC-AUC of 0.9289**, and a **Damaged class recall of 85.91%** (F1 = 0.85). An interactive Streamlit web application provides real-time single-image inference.

1. Problem Statement & Motivation

Following earthquakes and natural disasters, thousands of building facades, joints, columns, and walls require immediate structural screening. On-site assessment by licensed structural engineers requires weeks during catastrophe scenarios. Automated image-based damage classification provides rapid triage to prioritize emergency repairs and evacuation.

2. Dataset & Image Preprocessing

We utilize **PEER Hub ImageNet (Φ-Net) Task 2: Damage State** (Gao & Mosalam, 2018), cropped to 224×224×3:
• **Training Set:** 11,811 images (6,282 Damaged [53.2%], 5,529 Undamaged [46.8%]), stored as float32 arrays (~6.62 GiB).
• **Official Held-Out Test Set:** 1,460 images (745 Damaged [51.0%], 715 Undamaged [49.0%]), ~838 MiB.
• **Validation Split:** Stratified 80/20 train/val split (9,448 train / 2,363 val; seed 42). Contiguous block ordering requires strict stratified shuffling.
• **Preprocessing Inversion:** Dataset arrays have Caffe BGR ImageNet means [103.939, 116.779, 123.68] subtracted. We invert them to standard RGB in [0, 255] via: *RGB = clip(Image_BGR + μ_BGR, 0, 255)[::-1]*, then scale to [-1, 1] via *(RGB / 127.5) - 1.0* for MobileNetV2. Uploaded JPG/PNG photographs are standard RGB and bypass mean restoration.

3. Methodology & System Architecture

Due to physical CPU/RAM constraints (dual-core i3, 8 GB RAM), training full CNNs end-to-end causes system lockups. We formulate **transfer learning with disk-backed bottleneck caching**:
• **Backbone:** Pretrained MobileNetV2 (include_top=False, 2,257,984 frozen parameters in convolutional blocks).
• **Disk Caching:** Mini-batches of 8 are passed through the frozen extractor and GlobalAveragePooling2D, writing 1,280-dim feature vectors directly to disk via *np.lib.format.open_memmap* with atomic *.meta.json* progress sidecars. Working RAM remains below 1.5 GB.
• **Classification Head:** Input(1280) → Dropout(0.4) → Dense(64, ReLU) → Dense(2, Softmax), totaling 82,114 trainable parameters (2,340,098 total model parameters).
• **Training:** Adam (lr=0.001), Categorical Cross-Entropy, batch size 64, EarlyStopping (patience=5), ReduceLROnPlateau.

4. Experimental Results & Evaluation

The model converged in 19 epochs. Evaluation on the **1,460 official held-out test images** produced **84.25% accuracy** and **0.9289 ROC-AUC**.

Metric	Damaged (Class 0)	Undamaged (Class 1)	Macro Avg	Overall Test Set
Precision	**0.8366 (83.7%)**	0.8489 (84.9%)	0.8428 (84.3%)	0.8426 (84.3%)
Recall	**0.8591 (85.9%)**	0.8252 (82.5%)	0.8421 (84.2%)	0.8425 (84.2%)
F1-Score	**0.8477 (84.8%)**	0.8369 (83.7%)	0.8423 (84.2%)	0.8424 (84.2%)
Test Accuracy	—	—	—	**84.25% (1230 / 1460)**
ROC-AUC	**0.9289**	—	—	**0.9289**

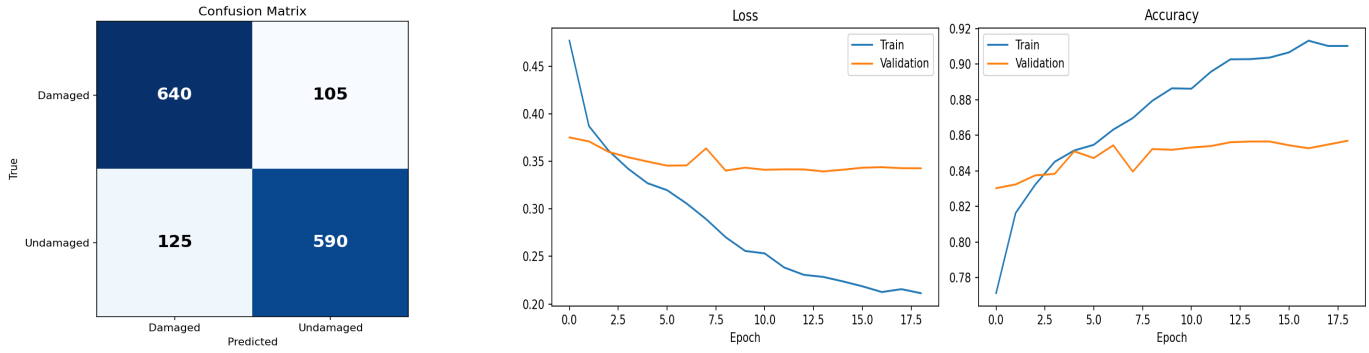


Figure 1: (Left) Official Test Set Confusion Matrix [640 TP, 590 TN, 125 FP, 105 FN]. (Right) Training and validation loss & accuracy curves over 19 epochs.

Confusion Matrix Analysis: Actual Damaged (745 total): 640 correct (TP), 105 missed (FN, 14.1% miss rate). Actual Undamaged (715 total): 590 correct (TN), 125 false alarms (FP, 17.5% alarm rate). In civil safety engineering, recall (85.91%) is prioritized to minimize dangerous structural oversights. Plausible visual failure modes for false negatives include low-contrast hairline fissures and shadow occlusions; plausible visual failure modes for false positives include high-contrast window frames, conduit lines, and architectural brick seams.

5. Deployment Prototype, Limitations & Future Work

• **Streamlit Web Application:** A functional UI (*app.py*) was built with resource-cached model loading, responsive single-photograph inference, confidence gauge, and civil engineering safety disclaimer.
• **Limitations:** Binary classification does not evaluate structural damage severity (e.g. slight vs. catastrophic) or identify failure modes (shear, flexure). Resolution is fixed at 224×224 px. Predictions must never replace certified civil engineering inspection.
• **Future Work:** Multi-class damage grading across Φ-Net Tasks 3–6, Grad-CAM saliency heatmaps, and edge-device mobile deployment.

References

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3. Sandler, M., et al. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. *CVPR 2018*, 4510–4520.
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